

Individual Report — OFDM Modulation and Demodulation Engine

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Project: Adaptive Self-Calibrating OFDM-Based Acoustic Communication

Code: ofdm/modem.py, ofdm/config.py · **Repository:** github.com/RukadeAkshay01/acoustic-ofdm

Stage: mid-term, approximately week 4 of 7

1. Scope of my work

I am responsible for the modem core: converting a bit stream into a waveform a speaker can play, and converting recorded samples back into a symbol grid. This covers QPSK mapping and demapping, the IFFT and cyclic prefix, the pilot and preamble structure, the peak-to-average power problem, and the shared parameter set the rest of the team builds against.

I do not own channel estimation, equalisation, synchronisation or the testing framework; those belong to Members 2 and 3, and their results are referenced here only where they were needed to verify my module.

2. What is implemented

The main design decision was to avoid a separate up-conversion stage. Instead of generating a complex baseband signal and mixing it to an audio carrier, I place the QPSK symbols directly on FFT bins 22–64 of a 256-point IFFT and enforce Hermitian symmetry, $X[N-k] = \text{conj}(X[k])$. A spectrum with this property has a purely real inverse transform, so the IFFT output is already a real audio signal in the 4–12 kHz band and can be written straight to a WAV file. I confirmed this numerically — the residual imaginary component after the transform is 7×10^{-17} , which is floating-point noise. This removes the carrier-phase recovery problem from the project entirely.

Parameter	Value	Reason
Sample rate	48 kHz	supported by consumer audio hardware without resampling
FFT size	256	187.5 Hz spacing; 6.67 ms symbol
Cyclic prefix	64 samples	1.33 ms guard, covering 0.46 m of excess path
Occupied band	4–12 kHz	laptop speakers roll off below, microphones above
Subcarriers	31 data + 12 pilot	pilot every 4th subcarrier
Modulation	QPSK, Gray	one symbol error costs one bit, not two
Raw payload rate	9 300 bit/s	62 bits per symbol

All of the above sit in a single configuration object with derived quantities as properties, so my teammates could change bandwidth or FFT size from one place. That turned out to matter later: restricting the modem to a calibrated subset of subcarriers became a one-line change rather than a modification to the modulator.

3. Peak-to-average power reduction

An OFDM symbol is the sum of many independent subcarriers, so its envelope is close to Gaussian and its peak-to-average power ratio is about 11 dB. A speaker is peak-limited, which means an unclipped waveform wastes most of the power the hardware could deliver, and received SNR follows average power. I implemented iterative clipping and filtering: clip to a set level above RMS, transform, zero everything outside the band, transform back, repeat.

Clipping limit	PAPR	Avg power	BER @ 8 dB	BER @ 14 dB
none	11.3 dB	+0.0 dB	1.9×10^{-4}	0
8 dB above RMS	9.2 dB	+2.1 dB	1.3×10^{-4}	0
6 dB above RMS	8.0 dB	+3.3 dB	3.2×10^{-4}	0
4 dB above RMS	7.1 dB	+4.2 dB	2.6×10^{-3}	5.4×10^{-5}

I selected 6 dB. It is the last setting that still reaches zero errors from 14 dB SNR upward while recovering 3.3 dB of transmit power. Clipping harder introduces an error floor that no amount of SNR removes, because the distortion then exceeds the noise.

4. Verification of my module

My module is verified in simulation only. The most useful check is that with the channel reduced to additive noise, the measured BER should lie on the coherent-QPSK theory curve — if the IFFT, cyclic prefix or demapper were wrong, it would not.

SNR	Measured BER	QPSK theory
0 dB	7.32×10^{-2}	7.86×10^{-2}
4 dB	1.20×10^{-2}	1.25×10^{-2}
8 dB	4.03×10^{-4}	1.91×10^{-4}

16 trials $\times$ 6 200 bits per point. The small excess at higher SNR is the in-band distortion from the clipping described in §3, which is a deliberate trade.

Other checks: Hermitian symmetry produces a real signal; an ideal loopback with no channel gives BER 0; and the configuration re-derives pilot, payload and bit-rate correctly when restricted to a subset of subcarriers.

5. A defect I introduced and corrected

My first version normalised the whole transmit frame — preamble and payload together — by a single global peak. This is wrong. The chirp preamble is a constant-envelope signal with roughly 3 dB PAPR while the OFDM block has about 9 dB, so dividing both by the same peak left the preamble far too quiet. The failure appeared in Member 3's synchroniser rather than in my module, which made it difficult to attribute: the correlator locked onto random payload correlation and the recovered BER was 0.49.

Normalising the preamble and the payload separately fixed it; detection confidence rose from a marginal $8.6\times$ the noise floor to $121\times$. The general lesson is that a single normalisation across signals with different crest factors is almost always wrong.

6. Honest assessment of progress

The modem core is functional and behaves correctly in simulation, but the project as a whole is at an early stage. I would put overall completion at roughly a third of what the proposal describes. The parts that are working are the parts that are easiest to verify on one machine; almost everything that makes this project *adaptive* and *self-calibrating* in the sense of the title is either untested on real hardware or not yet attempted.

Specifically, with respect to my own module:

- **All of my BER figures are from simulation.** Over-the-air results exist but are preliminary, and were obtained with one laptop transmitting to its own microphone — so the two ends share a clock, and the cross-device behaviour the proposal is built around is not demonstrated.
- **Fixed QPSK on every subcarrier is inadequate for real hardware.** Our measured speaker and microphone response varies by tens of dB across the band. The correct answer is bit-loading — more bits on strong subcarriers, fewer on weak ones — and I have not implemented it. The current workaround simply discards weak subcarriers, which wastes capacity.
- **The cyclic prefix length is unjustified.** 1.33 ms happens to cover the impulse response we measured on one desk, but we have not characterised a reverberant room, and I have no measurement that would justify the value in general.
- **No forward error correction in the modem.** A simple repetition code was added at the framing layer as a stopgap; it costs most of the throughput and is not a serious design.
- **Out-of-band emission is not controlled.** The cyclic-prefix discontinuity and the clipping both splatter energy outside 4–12 kHz. Windowed OFDM would address this and I have not implemented it.

7. Plan for the remaining weeks

- Implement bit-loading driven by the measured per-subcarrier SNR, which should remove the current error floor and recover throughput at the same time.
- Support the two-device test by making the transmitter configurable from a calibration set reported by the receiver.
- Measure the impulse response in a more reverberant environment and choose the cyclic prefix from that measurement rather than from convenience.
- Evaluate windowed OFDM for out-of-band suppression.

All figures in this report are generated from measured output and are reproducible with `python3 experiments/run_ber.py` and `python3 experiments/run_papr.py`. Nothing here is illustrative.